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Antenna and electronic device including the same

Abstract

According to various embodiments, an electronic device includes: a first housing including a first area, a second housing coupled to be slidable in a first direction from the first housing and including a second area overlapping the first area in a slide-in state, an antenna structure disposed in the first housing to overlap the first area when the first housing is viewed from the top, a conductive part disposed in the second area and electromagnetically connected to the antenna structure in the slide-in state, and wireless communication circuitry electrically connected to the antenna structure. The wireless communication circuitry may be configured to transmit and/or receive radio signals in at least one designated frequency band through the antenna structure and the conductive part in the slide-in state.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/511,910, filed Oct. 27, 2021 (U.S. Pat. No. 11,949,149), which is a Continuation of International Application No. PCT/KR2021/014098 filed Oct. 13, 2021, which claims priority to KR 10-2020-0143807 filed Oct. 30, 2020, the entire contents of which are all hereby incorporated herein by reference in their entireties.

BACKGROUND

Field

(1) The disclosure relates to an antenna and an electronic device including the same.

Description of Related Art

(2) An electronic device gradually becomes slim, and is developed to increase stiffness and enhance a design aspect and also to differentiate between functional elements thereof. The electronic device has its shape gradually changed into various shapes away from a uniform rectangular shape. The electronic device may have a deformable structure which can be easily carried and can also use a large-screen display. For example, as part of a deformable structure, an electronic device may have a structure (e.g., a rollable structure or a slidable structure) in which a display area is varied through at least two housings operating in a sliding way and a flexible display supported by the two housings. Such an electronic device may require an antenna capable of having excellent radiation performance regardless of a sliding operation and/or a rolling operation.

(3) An electronic device may include a deformable slidable electronic device (e.g., a rollable electronic device) capable of inducing the expansion of a display area when being used. The

slidable electronic device may include a first housing (e.g., a base housing, a base bracket or a base structure) and a second housing (e.g., a slide housing, a slide bracket or a slide structure) which may be coupled in a way to be movable to each other in at least partially fitted-together way. For example, the second housing may change a display area of a flexible display by being slid in or out in a designated direction and a designated distance from the first housing. The second housing may be coupled in a way to support at least a part of the flexible display (or an expandable display) and to operate at least partially in a sliding way from the first housing, and may induce a change in the display area by being manually slid in or out by a user or having its state automatically changed into a slide-in state or a slide-out state through an internal driving mechanism.

(4) The slidable electronic device may include at least one antenna disposed in the first housing including substantially most of electrical structures, such as a substrate, a sensor module or a battery. The at least one antenna may be configured to transmit or receive a radio signal in at least one designated frequency band by being electrically connected to wireless communication circuitry disposed in an internal space of the first housing and/or the substrate. The at least one antenna may include a conductive part disposed in a dielectric structure (e.g., an antenna carrier) disposed in the internal space of the first housing or formed through at least one non-conductive part in at least a part of an external appearance of the housing.

(5) However, in the slidable electronic device, if the second housing is disposed to be slidable from the first housing, the space in which the antenna is arranged may not be affordable due to a portion mutually overlapped in the slide-in state and an operating area of the flexible display. Alternatively, if at least one antenna is disposed in a corresponding area of the first housing overlapping the second housing in the slide-in state, radiation performance may deteriorate due to interference. Such a deterioration phenomenon may become more severe if a corresponding area of the second housing overlapping the first housing is made of a metal material.

SUMMARY

(6) Embodiments of the disclosure can provide an antenna having an improved degree of freedom in a part design because an antenna mounting space is secured regardless of an overlap area of two housings and an electronic device including the same.

(7) Embodiments of the disclosure can provide an electronic device including an antenna having designated radiation performance regardless of a slide-in/slide-out operation.

(8) According to various example embodiments, an electronic device includes: a first housing including a first area, a second housing coupled to be slidable in a first direction from the first housing and including a second area overlapping the first area in a slide-in state, an antenna structure disposed in the first housing to overlap the first area when the first housing is viewed from the top, a conductive part disposed in the second area and electromagnetically connected to the antenna structure in the slide-in state, and wireless communication circuitry electrically connected to the antenna structure. The wireless communication circuitry may be configured to transmit and/or receive radio signals in at least one designated frequency band through the antenna structure and the conductive part in the slide-in state.

(9) The electronic device according to various example embodiments of the present disclosure can provide an electrical connection structure which enables a surrounding conductive structure to operate as a part of the antenna by electrically connecting, to the antenna, the surrounding conductive structure overlapping the antenna in a slide-in operation. Accordingly, the antenna can maintain stable radiation performance regardless of a sliding operation.

(10) In addition, various effects directly or indirectly understood through this document may be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a block diagram of illustrating an example electronic device in a network environment according to various embodiments;
- (3) FIGS. 2A and 2B are diagrams illustrating the front of an electronic device in a slide-in state and a slide-out state, respectively, according to various embodiments;
- (4) FIGS. 3A and 3B are diagrams illustrating the rear of the electronic device in the slide-in state and the slide-out state, respectively, according to various embodiments;
- (5) FIG. 4 is an exploded perspective view of the electronic device according to various embodiments;
- (6) FIGS. 5A and 5B are cross-sectional views of the electronic device in the slide-in state and slide-out state of the electronic device, respectively, according to various embodiments;
- (7) FIG. 6A is a perspective view of a part of the electronic device including an antenna structure and a conductive part according to various embodiments;
- (8) FIG. 6B is a partial perspective view illustrating the state in which the antenna structure and the conductive part have been electrically connected through a conductive connection member according to various embodiments;
- (9) FIG. 7 is a graph comparing radiation performance of antennas depending on the presence or absence of the conductive part in FIG. 6A according to various embodiments;
- (10) FIGS. 8A and 8B are diagrams illustrating a part of the electronic device including an antenna structure and a conductive part according to various embodiments;
- (11) FIG. 9 is a graph illustrating radiation performance of an antenna in the configurations of FIGS. 8A and 8B according to various embodiments;
- (12) FIGS. 10A and 10B are diagrams illustrating a part of the electronic device including an antenna structure and a conductive part according to various embodiments;
- (13) FIG. 11 is a graph illustrating radiation performance of an antenna in the configurations of FIGS. 10A and 10B according to various embodiments; and
- (14) FIG. 12 is a diagram illustrating the electronic device including a plurality of antenna structures and a plurality of conductive parts according to various embodiments.
- (15) In relation to the description of the drawings, the same or similar reference numerals may be used with respect to the same or similar constituent elements.

DETAILED DESCRIPTION

- (16) FIG. 1 is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments.
- (17) Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In various embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the

antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(18) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(19) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(20) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(21) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(22) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(23) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may

be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(24) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(25) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(26) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(27) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(28) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(29) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(30) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(31) The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(32) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(33) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular

communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(34) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(35) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(36) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(37) At least some of the above-described components may be coupled mutually and communicate

signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(38) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(39) FIGS. 2A and 2B are diagrams illustrating the front of an electronic device in a slide-in state and a slide-out state, respectively, according to various embodiments. FIGS. 3A and 3B are diagrams illustrating the rear of the electronic device in the slide-in state and the slide-out state, respectively, according to various embodiments.

(40) Referring to FIGS. 2A, 2B, 3A and 3B, the electronic device **200** may include a first housing **210** (e.g., a base housing), a second housing **220** (e.g., a slide housing) coupled to be movable from the first housing **210** in a designated direction (e.g., an X axis direction) within a designated round trip, and a flexible display **230** (e.g., an expandable display) disposed to be supported through at least a part of the first housing **210** and the second housing **220**. According to an embodiment, the electronic device **200** may include a bendable member (or a bendable support member) (e.g., a bendable member **240** in FIG. 5A) (e.g., a hinge rail, bendable support or a multi-joint hinge module) which forms the same plane along with at least a part of the first housing **210** in the slide-out state and is received in an internal space (e.g., a second space **2201** in FIG. 5A) of the second housing **220** in the slide-in state. According to an embodiment, at least a part of the flexible display **230** may be disposed to be not seen from the outside by being received in the internal space (e.g., the second space **2201** in FIG. 5A) of the second housing **220** while being supported by a bendable member (e.g., the bendable member **240** in FIG. 5A) in the slide-in state. According to an embodiment, at least a part of the flexible display **230** may be disposed to be seen from the outside while being supported by a bendable member (e.g., the bendable member **240** in FIG. 5A) that forms at least partially the same plane along with the first housing **210** in the slide-out state.

(41) According to various embodiments, the electronic device **200** may include a front **200a** (e.g., a first face), a rear **200b** (e.g., a second face) facing a direction opposite to the front **200a**, and a side (not illustrated) surrounding the space between the front **200a** and the rear **200b**. According to an embodiment, the electronic device **200** may include the first housing **210** including a first side member **211** and the second housing **220** including a second side member **221**. According to an

embodiment, the first side member **211** may include a first side **2111** having a first length in a first direction (the X axis direction), a second side **2112** extended to have a second length longer than the first length in a direction substantially perpendicular to the first side **2111**, and a third side **2113** extended from the second side **2112** in substantially parallel to the first side **2111** and having the first length. According to an embodiment, the first side member **211** may be made of at least partially a conductive material (e.g., metal). According to an embodiment, at least a part of the first side member **211** may include a first support member **212** extended up to at least a part of an internal space (e.g., a first space **2101** in FIG. 5A) of the first housing **210**.

(42) According to various embodiments, the second side member **221** may include a fourth side **2211** facing the first side **2111** and having a third length, a fifth side **2212** extending from the fourth side **2211** in a direction substantially parallel to the second side **2112** and having a fourth length longer than the third length, and a sixth side **2213** extending from the fifth side **2212** substantially in parallel to the third side **2113** and having the third length. According to an embodiment, the second side member **221** may be made of at least partially a conductive material (e.g., metal). According to an embodiment, at least a part of the second side member **221** may include a second support member **222** extended up to at least a part of an internal space (e.g., the second space **2201** in FIG. 5A) of the second housing **220**. According to an embodiment, each of the first side **2111** and the fourth side **2211**, and the third side **2113** and the sixth side **2213** may be coupled to be mutually slidable. According to an embodiment, in the slide-in state, at least a part of the first side **2111** may be disposed to overlap at least a part of the fourth side **2211**. According to an embodiment, in the slide-in state, at least a part of the third side **2113** may be disposed to overlap at least a part of the sixth side **2213**. According to an embodiment, in the slide-in state, at least a part of the first support member **212** of the first side member **211** may be disposed to overlap the second support member **222** of the second side member **221**. In an embodiment, the first support member **212** may include a non-overlap part **212a** not overlapping the second support member **222** in the slide-in state. In an embodiment, in the slide-in state, the non-overlap part **212a** may be formed separately from the first support member **212** and coupled without overlapping the second support member **222**. According to an embodiment, the non-overlap part **212a** may be used as an area where electronic parts (e.g., the camera module **216**, the sensor module **217** or the flash **218**) are disposed. According to an embodiment, the area of the front **200a** and rear **200b** of the electronic device **200** may be varied depending on the slide-in state and slide-out state of the electronic device **200**. In an embodiment, the electronic device **200** may include a first rear cover (not illustrated) disposed in at least a part of the first housing **210** and a second rear cover (not illustrated) disposed in at least a part of the second housing **220** in the rear **200b**. In such a case, the first rear cover and/or the second rear cover may be integrated and formed with the side members **211** and **221**, respectively. According to an embodiment, the first rear cover and/or the second rear cover may be made of polymer, coating or colored glass, ceramic, metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two of the materials. In an embodiment, the first rear cover and/or the second rear cover may be extended up to at least a part of the side members **211** and **221**, respectively. In an embodiment, the first support member **212** may be substituted with the first rear cover. The second support member **222** may be substituted with the second rear cover.

(43) According to various embodiments, the electronic device **200** may include the flexible display **230** disposed to be supported by at least a part of the first housing **210** and the second housing **220**. According to an embodiment, the flexible display **230** may include a first part **230a** (e.g., a plane unit) always seen from the outside and a second part **230b** (e.g., a bendable part) extended from the first part **230a** and slid in an internal space (e.g., the first space **2201** in FIG. 5A) of the second housing **220** in a way to be not seen from the outside in the slide-in state. According to an embodiment, the first part **230a** may be disposed to be supported by the first housing **210**. The second part **230b** may be disposed to be supported by a bendable member (e.g., the bendable member **240** in FIG. 5A). According to an embodiment, the second part **230b** of the flexible display

230 may be disposed to slide in an internal space (e.g., the second space **2201** in FIG. 5A) of the second housing **220** and to be not exposed to the outside in the slide-in state of the electronic device **200**. The second part **230b** may be extended from the first part **230a** while being supported by a bendable member (e.g., the bendable member **240** in FIG. 5A) and may be exposed to the outside in a way to form substantially the same plane as the first part **230a**, in the state in which the electronic device **200** has slid out in the designated first direction (a direction {circle around (1)}). Accordingly, the display area of the flexible display **230** of the electronic device **200** may be varied as the second housing **220** moves in the designated first direction (e.g., the direction {circle around (1)}) from the first housing **210**.

(44) According to various embodiments, the first housing **210** may be slidably coupled in a way to be at least partially slid out from the second housing **220** or slid in at least a part of the second housing **220**. According to an embodiment, the electronic device **200** may be configured to have a first width **W1** from the second side **2112** to the fourth side **2212** in the slide-in state. According to an embodiment, the electronic device **200** may be configured to have a third width **W3** greater than the first width **W1** because a bendable member (e.g., the bendable member **240** in FIG. 5A) slid in an internal space (e.g., the second space **2201** in FIG. 5A) of the second housing **210** is moved to have an additional second width **W2** in the slide-out state. For example, the flexible display **230** may have a display area substantially having the first width **W1** in the slide-in state, and may have an extended display area substantially having the third width **W3** in the slide-out state.

(45) According to various embodiments, the second housing **220** may be operated through a manipulation of a user. For example, a state of the electronic device **200** may change into the slide-out state through a manipulation of a user who presses an external surface of the flexible display **230** in the designated first direction (e.g., the direction {circle around (1)}) in the slide-in state. In an embodiment, the second housing **220** of the electronic device **200** may be slid out in the designated first direction (e.g., the direction {circle around (1)}) through a manipulation of a button (not illustrated) of a locker (not illustrated) exposed to the outside. In an embodiment, the second housing **220** may be automatically operated through a driving mechanism (e.g., a driving motor, a deceleration module and/or a gear assembly) disposed in an internal space (e.g., the first space **2101** in FIG. 5A) of the first housing **210** and/or an internal space (e.g., the second space **2201** in FIG. 5A) of the second housing **220**. According to an embodiment, the electronic device **200** may be configured to control an operation of the second housing **220** through the driving mechanism when detecting an event for a change into the slide-in/slide-out state of the electronic device **200** through a processor (e.g., the processor **120** in FIG. 1). In an embodiment, a processor (e.g., the processor **120** in FIG. 1) of the electronic device **200** may control the flexible display **230** to display an object in various ways and to execute an application program in accordance with a changed display area of the flexible display **230** in the slide-in state, the slide-out state or an intermediate state.

(46) According to various embodiments, the electronic device **200** may include at least one of an input device (not illustrated), a sound output device **206**, sensor modules **204** and **217**, camera modules **205** and **216**, a connector port (not illustrated), a key input device (not illustrated) or an indicator (not illustrated). In an embodiment, the electronic device **200** may be configured to omit at least one of the aforementioned elements or to additionally include other elements.

(47) According to various embodiments, the input device may include a microphone. In an embodiment, the input device may include a plurality of microphones disposed to be capable of detecting the direction of a sound. The sound output device **206** may include a speaker. The sound output device **206** may include a receiver for communication. In an embodiment, the sound output device **206** may include an external speaker. In an embodiment, the sound output device **206** may include an operating speaker (e.g., a piezo speaker) without a separate speaker hole.

(48) According to various embodiments, the sensor modules **204** and **217** may generate an electric signal or a data value corresponding to an operating state within the electronic device **200** or an

external environment state. The sensor modules **204** and **217** may include a first sensor module **204** (e.g., a proximity sensor or an illuminance sensor) disposed in the front **200a** of the electronic device **200** and/or a second sensor module **217** (e.g., a heart rate monitoring (HRM) sensor) disposed in the rear **200b** thereof, for example. According to an embodiment, the first sensor module **204** may be disposed under the flexible display **230** in the front **200a** of the electronic device **200**. According to an embodiment, the first sensor module **204** and/or the second sensor module **217** may include at least one of a proximity sensor, an illuminance sensor, a time of flight (TOF) sensor, an ultrasonic sensor, a fingerprint recognition sensor, a gesture sensor, a gyro sensor, an atmosphere sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a bio sensor, a temperature sensor or a humidity sensor.

(49) According to various embodiments, the camera modules **205** and **216** may include a first camera module **205** disposed in the front **200a** of the electronic device **200** and a second camera module **216** disposed in the rear **200b** thereof. According to an embodiment, the electronic device **200** may include a flash **218** disposed near the second camera module **216**. According to an embodiment, the camera modules **205** and **216** may include one or a plurality of lenses, an image sensor and/or an image signal processor. According to an embodiment, the first camera module **205** may be disposed under the flexible display **230** and configured to photograph a subject through a part of an activation area of the flexible display **230**. According to an embodiment, the flash **218** may include a light-emitting diode or a xenon lamp, for example.

(50) According to various embodiments, the first camera module **205** of the camera modules **205** and **216** and some sensor modules **204** of the sensor modules **204** and **217** may be disposed to detect an external environment through the flexible display **230**. For example, the first camera module **205** or some sensor modules **204** may be disposed to neighbor an external environment through an opening or a transmission area perforated in the flexible display **230** in an internal space of the electronic device **200**. According to an embodiment, an area that faces the first camera module **205** of the flexible display **230** may be formed as a transmission area having designated transmittance as a part of an area where content is displayed. According to an embodiment, the transmission area may be formed to have transmittance having a range of about 5% to about 20%. Such a transmission area may include an area through which light for generating an image by being focused by the image sensor passes and which overlaps a valid area (e.g., a view angle area) of the first camera module **205**. For example, the transmission area of the flexible display **230** may include an area in which the density and/or wiring density of pixels is lower than that of surrounding pixels. For example, the transmission area may substitute the aforementioned opening. For example, some camera modules **205** may include an under display camera (UDC). In an embodiment, some sensor modules **204** may be disposed to perform functions thereof without being visually exposed through the flexible display **230** in an internal space of the electronic device **200**.

(51) According to various embodiments, the electronic device **200** may include at least one antenna structure **213** (e.g., at least one antenna) disposed in the first housing **210** and configured to transmit and/or receive a radio signal in at least one frequency band. According to an embodiment, the at least one antenna structure **213** may be disposed in a part corresponding to a first area **R1** included in at least a part of the first support member **212** of the first housing **210**. According to an embodiment, the first area **R1** may be disposed in an area corresponding to a non-conductive part **2131** (e.g., conductor) disposed in at least some area of the first support member **212**. According to an embodiment, the second housing **220** may include a conductive part **223** overlapped with the first area **R1** and disposed in the second area **R2** of the second support member **222** in the slide-in state. According to an embodiment, the conductive part **223** may be disposed through a non-conductive part **2231** of the second support member. According to an embodiment, the conductive part **223** may be disposed to be electromagnetically connected to the antenna structure **213** in the slide-in state. In an embodiment, the electronic device **200** may further include a conductive pad

214 electrically connected to the antenna structure **213** so as to be electromagnetically connected to the conductive part **223** in the slide-in state if the antenna structure **213** is disposed in the first housing **210**. In an embodiment, the conductive part **223** may be disposed to be divided through the at least one non-conductive part **2231** in the second support member **222**. In an embodiment, in the slide-in state, the electromagnetic connection structure of the antenna structure **213** and the conductive part **223** may be disposed in at least one overlap area that overlaps each other among sides (e.g., between the first side **2111** and the fourth side **2211** and/or between the third side **2113** and the sixth side **2213**) of the electronic device **200**.

(52) FIG. **4** is an exploded perspective view of the electronic device according to various embodiments.

(53) Referring to FIG. **4**, the electronic device **200** may include the first housing **210** including a first space (e.g., the first space **2101** in FIG. **5A**), the second housing **220** slidably coupled to the first housing **210** and including a second space (e.g., the second space **2201** in FIG. **5A**), the bendable member **240** rotatably disposed in a second space (e.g., the second space **2201** in FIG. **5A**), and the flexible display **230** disposed to be supported by the bendable member **240** and the first housing **210**. According to an embodiment, a first space (e.g., the first space **2201** in FIG. **5A**) of the first housing **210** may be provided through the coupling of a first bracket housing **210a** and a second bracket housing **210b**. In an embodiment, at least a part of the first bracket housing **210a** may include the first support member **212** or may be substituted with the first support member **212**. According to an embodiment, the electronic device **200** may include a substrate **250** disposed in a first space (e.g., the first space **2201** in FIG. **5A**). According to an embodiment, the substrate **250** may include wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**). According to an embodiment, the electronic device **200** may include a camera module (e.g., the camera module **216** in FIG. **3A**) or a sensor module (e.g., the sensor module **217** in FIG. **3A**) disposed in the substrate **250** in a first space (e.g., the first space **2101** in FIG. **5A**). According to an embodiment, the bendable member **240** may be disposed to have one end fixed to the first housing **210** and the other end rotatably received in a second space (e.g., the second space **2201** in FIG. **5A**) of the second housing **220**. For example, the bendable member **240** may be at least partially received in a second space (e.g., the second space **2201** in FIG. **5A** in the slide-in state, and may be slid out from a second space (e.g., the second space **2201** in FIG. **5A**) in a way to form substantially the same plane as the first housing **210** in the slide-out state. Accordingly, the display area of the flexible display **230** supported by the first housing **210** and the bendable member **240** may be varied depending on a sliding operation. According to an embodiment, the electronic device **200** may further include a guide rail **242** disposed on the side of the coupled first bracket housing **210a** and second bracket housing **210b**, for being guided into an internal space (e.g., the second space **2201** in FIG. **5A**) of the second housing **220**. In an embodiment, the electronic device **200** may further include a cover member **222a** disposed to cover the second support member **222** of the second housing **220**.

(54) According to various embodiments, the electronic device **200** may include the antenna structure **213** disposed in a first space (e.g., the first space **2201** in FIG. **5A**) of the first housing **210**. According to an embodiment, the antenna structure **213** may be disposed through a dielectric structure **2132** (e.g., an antenna carrier). In such a case, the antenna structure **213** may include a laser direct structuring (LDS) pattern. In an embodiment, the antenna structure **213** may include a conductive pattern directly disposed on the substrate **250**. In an embodiment, the antenna structure **213** may include at least one conductive part divided through at least one non-conductive part in the first support member **212**. According to an embodiment, the dielectric structure **2132** including the antenna structure **213** may be disposed at a location corresponding to the non-conductive part **2131** formed in the first support member **212** of the first housing **210**, and may be electrically connected to the conductive pad **214** disposed in the first support member **212**. According to an embodiment, the second housing **220** may include the conductive part **223** formed to be divided

through the non-conductive part **2231** in the second support member **222**. For example, in the slide-in state, at least a part of the conductive part **223** is disposed to overlap the antenna structure **213** and electromagnetically connected through the conductive pad **214**, thereby being capable of reducing the deterioration of radiation performance of the antenna structure **213** attributable to interference of the second housing **220**. According to an embodiment, the electronic device **200** may further include a matching circuit (not illustrated) in order to transmit and/or receive a signal having a designated frequency band using the antenna structure **213** in the slide-in state or the slide-out state. For example, the matching circuit may include at least one of an inductor, a capacitor or a switch.

(55) FIGS. **5A** and **5B** are cross-sectional views of the electronic device in the slide-in state and slide-out state, respectively, of the electronic device according to various embodiments.

(56) Referring to FIGS. **5A** and **5B**, the electronic device **200** may include the first housing **210** including the first space **2101**, the second housing **220** including the second space **2201** disposed to be slidable in a designated direction (the X axis direction) within a designated round trip from the first housing **210**, the bendable member **240** coupled to the first housing **210** and disposed to be supported through a support **241** (e.g., a support roller) disposed in the second housing **220**, and the flexible display **230** disposed to be supported by the first housing **210** and the bendable member **240**. According to an embodiment, the flexible display **230** may be disposed to have a first display area (e.g., an area corresponding to the first width **W1** in FIG. **2B**) by being at least partially slid in the second space **2201** of the second housing **220** through the support of the bendable member **240** in the slide-in state. According to an embodiment, the flexible display **230** may be disposed to have a second display area (e.g., an area corresponding to the third width **W3** in FIG. **2B**) greater than the first display area by being at least partially slid out from the second space **2201** to the outside through the support of the bendable member **240** in the slide-out state.

(57) According to various embodiments, the first housing **210** may include the antenna structure **213** disposed in the first space **2101**. According to an embodiment, the antenna structure **213** may be disposed in the dielectric structure **2132** (e.g., an antenna carrier). In an embodiment, the antenna structure **213** may include a conductive pattern formed on the substrate **250** disposed in the first space **2101**. In an embodiment, the antenna structure **213** may include at least a part of the first support member **212** divided through the at least one non-conductive part **2131**. According to an embodiment, the antenna structure **213** may be electrically connected to wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**) disposed in the substrate **250**. In an embodiment, wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**) may be disposed at a location isolated from the substrate **250** in the first space **2101** of the first housing **210**, and may be electrically connected to the substrate through an electrical connection member (e.g., an RF cable). According to an embodiment, the antenna structure **213** may be electrically connected to the conductive pad **214** of the first support member **212**. In an embodiment, the conductive pad **214** may be divided through the non-conductive part **2131**, and may be substituted with a conductive part disposed as a part of the first support member **212**. According to an embodiment, when the first support member **212** is viewed from the top, an area of the first support member **212** at least partially overlapping the antenna structure **213** may help to maintain radiation performance of the antenna structure **213** because the area is formed of the non-conductive part **2131**. In an embodiment, the non-conductive part **2131** may be formed to include at least a part of the non-overlap part **212a** in addition to a part covered with the second support member **222** of the second housing **220** in the slide-in state.

(58) According to various embodiments, the second housing **220** may include the conductive part **223** disposed in at least a part of the second support member **222** divided through the at least one non-conductive part **2231**. According to an embodiment, the conductive part **223** may be disposed at a location at least partially overlapping the antenna structure **213** in the slide-in state when the second support member **222** is viewed from the top. According to an embodiment, the conductive

part **223** may be electrically connected to the conductive pad **214** through a conductive connection member **224** disposed in the first space **2201** of the first housing **210** in the slide-in state. According to an embodiment, the conductive connection member **224** may be fixed to be electrically connected to the conductive part **223** in the internal space **2201** of the second housing **220**. In an embodiment, the conductive connection member **224** may be fixed to the first housing **210** in a way to be electrically connected to the conductive pad **214**. In such a case, the conductive connection member **224** may be electrically connected to the conductive part **223** of the second housing **220** in the slide-in state. According to an embodiment, the antenna structure **213** is electrically connected to the conductive part **223** of the second housing **220** through the conductive pad **214** and the conductive connection member **224** in the slide-in state, so that the deterioration of radiation performance attributable to interference of the second housing **220** can be reduced. According to an embodiment, the antenna structure **213** can maintain the state in which the antenna structure **213** has been electrically disconnected from the conductive part **223** because the conductive connection member **224** is isolated from the conductive pad **214** in the slide-out state. The conductive connection member **224** may include a C-clip or a pogo pin as a connection member having elasticity, for example.

(59) According to various embodiments, wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**) may be configured to transmit and/or receive radio signals in at least one designated frequency band through the antenna structure **213** and the conductive part **223** in the slide-in state. According to an embodiment, wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**) may be configured to transmit and/or receive radio signals in at least one designated frequency band through only the antenna structure **213** in the slide-out state. According to an embodiment, wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**) may be configured to transmit and/or receive radio signals in at least one frequency band of a low band (e.g., about 700 MHz to 900 MHz), a mid band (about 1700 MHz to 2100 MHz), a high band (about 2300 MHz to 2700 MHz) or a sub-6 band (about 3 GHz to 6 GHz) through the antenna structure **213** and/or the antenna structure **213** and the conductive part **223**.

(60) FIG. **6A** is a perspective view illustrating a part of the electronic device including an antenna structure and a conductive part according to various embodiments. FIG. **6B** is a partial perspective view schematically illustrating the state in which the antenna structure and the conductive part have been electrically connected through the conductive connection member according to various embodiments.

(61) Referring to FIGS. **6A** and **6B**, the electronic device **200** may include the first housing **210** and the second housing **220** coupled in a way to be slidable in a designated direction (e.g., the direction {circle around (1)}) from the first housing **210**). According to an embodiment, the first housing **210** may include the first support member **212**. According to an embodiment, the second housing **220** may include the second support member **222**. According to an embodiment, the electronic device **200** may operate in such a way that at least a part of the first support member **212** is overlapped to be not seen from the outside by the second support member **222** in the slide-in state. According to an embodiment, the electronic device **200** may include the antenna structure **213** overlapped with the second support member **222** and disposed in a corresponding area of the first housing **210**, when the first support member **212** is viewed from the top in the slide-in state. According to an embodiment, the antenna structure **213** may be formed through a conductive part divided through the non-conductive part **2131** in the first support member **212**. According to an embodiment, the electronic device **200** may include the conductive part **223** at least partially overlapped with the antenna structure **213** and disposed in a corresponding area of the second support member **222**, when the second support member **212** is viewed from the top in the slide-in state. According to an embodiment, the conductive part **223** may be disposed to be divided through the at least one non-conductive part **2231** disposed in the second support member **222**.

(62) According to various embodiments, the electronic device **200** may include the conductive connection member **224** for electrically and physically connecting the antenna structure **213** and the conductive part **223** in the slide-in state. According to an embodiment, the conductive connection member **224** may be disposed to be not seen from the outside between the first support member **212** and the second support member **222** in the slide-in state. In an embodiment, the conductive connection member **224** may be disposed to be not seen from the outside in the slide-out state. According to an embodiment, the conductive connection member **224** may be disposed in the second housing **220**. According to an embodiment, the conductive connection member **224** may include at least one of a conductive spring, a conductive tape, a pogo pin or a conductive C-clip. According to an embodiment, the conductive connection member **224** may be fixed to the conductive part **223** through at least one of ultrasonic welding, conductive bonding, conductive taping or structural coupling.

(63) FIG. 7 is a graph comparing radiation performance of antennas depending on the presence or absence of the conductive part in FIG. 6A according to various embodiments.

(64) From FIGS. 6A, 6B and 7, it may be seen that radiation performance of the antenna structure **213** (e.g., an antenna) is suddenly decreased due to interference of the second housing **220** in the slide-in state in designated frequency bands (e.g., a high band (an area **701**) and a sub-6 band (an area **702**)) compared to the slide-out state (a graph **712**) when the conductive part **223** is not present (a graph **711**), whereas the antenna structure **213** according to an example embodiment of the present disclosure maintains a gain of 10 dB or more similar to the slide-out state (the graph **712**) when the antenna structure **213** is electrically connected to the conductive part **223** (the graph **713**) in the slide-in state. This may mean that the deterioration of radiation performance is reduced although the antenna structure **213** overlaps at least a part of the second housing **220** when being electromagnetically connected to the conductive part **223** in the slide-in state.

(65) FIGS. 8A and 8B are diagrams illustrating a part of the electronic device including an antenna structure and a conductive part according to various embodiments.

(66) In describing the electronic device **200** of FIGS. 8A and 8B, substantially the same elements as those of the electronic device **200** of FIGS. 6A and 6B have been assigned the same reference numerals, and a detailed description thereof may not be repeated here.

(67) Referring to FIGS. 8A and 8B, the electronic device **200** may include a first housing **210** and a second housing **220** coupled in a way to be slidable in a designated direction (e.g., the direction {circle around (1)}) from the first housing **210**. According to an embodiment, the first housing **210** may include a first support member **212**. For example, the first support member **212** may be extended from a side to an internal space. According to an embodiment, the second housing **220** may include a second support member **222**. For example, the second support member **222** may be extended from a side to an internal space. According to an embodiment, the electronic device **200** may operate in such a way that at least a part of the first support member **212** is overlapped to be not seen from the outside by the second support member **222** in the slide-in state. According to an embodiment, the electronic device **200** may include an antenna structure **215** overlapped with at least a part of the second support member **222** and disposed in a corresponding area of the first housing **210** when the first support member **212** is viewed from the top in the slide-in state. In an embodiment, the antenna structure **215** may be disposed within the first housing **210** or may be disposed to include a part divided through a non-conductive part in the first support member **212** made of a conductive material.

(68) According to various embodiments, the electronic device **200** may include a conductive part **225** disposed to at least partially overlap the antenna structure **215** when the second support member **222** is viewed from the top in the slide-in state. According to an embodiment, the conductive part **225** may be formed to have a designated shape through at least one non-conductive part **2231** in the second support member **222**. According to an embodiment, the conductive part **225** may be formed to be extended from the second support member **222** made of a conductive material

and to have a slit **2251**. According to an embodiment, the slit **2251** may be formed to have a length substantially in the same direction as the direction of a length in which an antenna structure **215** is formed. In an embodiment, the antenna structure **215** may be disposed to be electrically connected to the conductive part **225** through a conductive connection member (e.g., the conductive connection member **224** in FIG. 7B) in the slide-in state.

(69) FIG. 9 is a graph illustrating radiation performance of the antenna in the configurations of FIGS. 8A and 8B according to various embodiments.

(70) From FIGS. 8A, 8B and 9, it may be seen that when overlapping the conductive part **225** (a graph **911**) in the slide-in state in a designated frequency band (e.g., a sub-6 band (an area **901**)), the antenna structure **215** (e.g., an antenna) maintains a gain of 10 dB or more similar to the slide-out state (a graph **912**). This may mean that although the antenna structure **215** overlaps at least a part of the second housing **220** when electromagnetically connected to the conductive part **225** in the slide-in state, the deterioration of radiation performance is reduced because designated performance is generated as in a case where the antenna structure **215** solely operates.

(71) FIGS. 10A and 10B are diagrams illustrating a part of the electronic device including an antenna structure and a conductive part according to various embodiments.

(72) In describing the electronic device **200** of FIGS. 10A and 10B, substantially the same elements as those of the electronic device **200** of FIGS. 8A and 8B have been assigned the same reference numerals, and a detailed description thereof may not be repeated here.

(73) Referring to FIGS. 10A and 10B, the electronic device **200** may include a first housing **210** and a second housing **220** coupled in a way to be slidable in a designated direction (e.g., the direction {circle around (1)}) from the first housing **210**. According to an embodiment, the first housing **210** may include a first support member **212**. According to an embodiment, the second housing **220** may include a second support member **222**. According to an embodiment, the electronic device **200** may operate in such a way that at least a part of the first support member **212** is overlapped to be not seen from the outside by the second support member **222** in the slide-in state. According to an embodiment, the electronic device **200** may include an antenna structure **216** overlapped with at least a part of the second support member **222** and disposed in a corresponding area of the first housing **210** when the first support member **212** is viewed from the top in the slide-in state. According to an embodiment, the electronic device **200** may include a conductive part **225** disposed through a slit **2251** in a way to at least partially overlap the antenna structure **216** when the second support member **222** is viewed from the top in the slide-in state.

(74) According to various embodiments, the conductive part **225** may be disposed at a location capacitively coupled to the antenna structure **216** in the slide-in state. For example, the conductive part **225** may be disposed to have a designated gap “g” through which the conductive part **225** may be capacitively coupled to the antenna structure **216** in the slide-in state. In such a case, it may be advantageous that the antenna structure **216** is formed to be divided through at least one non-conductive part **2131** in the first support member **212** made of a conductive material. According to an embodiment, the conductive part **225** may be adjacent to the antenna structure **216** in a way to be capacitively coupled thereto in a direction parallel and/or perpendicular to a slide-in/slide-out direction (e.g., the direction {circle around (1)}) in the slide-in state.

(75) FIG. 11 is a graph illustrating radiation performance of the antenna in the configurations of FIGS. 10A and 10B according to various embodiments.

(76) From FIGS. 10A, 10B and 11, it may be seen that when being adjacent to the conductive part **225** in a way to be capacitively coupled thereto (a graph **1111**) in the slide-in state in designated frequency bands (e.g., a high band (an area **1101**) and a sub-6 band (an area **1102**)), the antenna structure **216** (e.g., an antenna) maintains a gain of 10 dB or more similar to the slide-out state (a graph **1112**). This may mean that the deterioration of radiation performance is reduced because designated performance is generated as in a case where the antenna structure **216** solely operates although the antenna structure **216** overlaps at least a part of the second housing **220** when

electrically connected to the conductive part **225** in a way to be capacitively coupled thereto in the slide-in state.

(77) FIG. **12** is a diagram illustrating an electronic device including a plurality of antenna structures and a plurality of conductive parts according to various embodiments.

(78) Referring to FIG. **12**, an electronic device **200** may include a first housing **210** and a second housing **220** coupled in a way to be slidable in a designated direction (e.g., the direction {circle around (1)}) from the first housing **210**. According to an embodiment, the first housing **210** may include a first support member **212**. According to an embodiment, the second housing **220** may include a second support member **222**. According to an embodiment, the electronic device **200** may operate in such a way that at least a part of the first support member **212** is overlapped to be not seen from the outside by the second support member **222** in the slide-in state. According to an embodiment, the electronic device **200** may include antennas **A1**, **A2**, and **A3** overlapped with at least a part of the second support member **222** and disposed in a corresponding area of the first housing **210** when the first support member **212** is viewed from the top in the slide-in state.

According to an embodiment, the antennas **A1**, **A2**, and **A3** may be formed through a first antenna structure **213**, a second antenna structure **215** and/or a third antenna structure **217** disposed to be divided through a non-conductive part **2131** in the first support member **212** made of a conductive material.

(79) According to various embodiments, the electronic device **200** may include conductive parts **223**, **225**, and **227** disposed to at least partially overlap the antenna structures **213**, **215**, and **217**, respectively, when the first support member **212** is viewed from the top in the slide-in state. According to an embodiment, the conductive parts **223**, **225**, and **227** may be disposed to be divided or extended through at least one non-conductive part **2231** and **2232** in the second support member **222** made of a conductive material. In an embodiment, the conductive part **225**, that is, any one of the conductive parts **223**, **225**, and **227**, may be disposed to include a slit **2251** formed through the non-conductive part **2231**. According to an embodiment, the deterioration of radiation performance of the antennas **A1**, **A2**, and **A3** attributable to interference of the second housing **220** can be reduced through the conductive parts **223**, **225**, and **227** disposed to overlap the antenna structures **213**, **215**, and **217** in the slide-in state. In an embodiment, the antennas **A1**, **A2**, and **A3** may be configured to operate in the same or different frequency bands. In an embodiment, two or more antennas may be formed in the electronic device **200**.

(80) According to various example embodiments, an electronic device (e.g., the electronic device **200** in FIG. **3B**) includes: a first housing (e.g., the first housing **210** in FIG. **3B**) including a first area (e.g., the first area **R1** in FIG. **3B**), a second housing (e.g., the second housing **220** in FIG. **3B**) coupled to be slidable in a first direction (e.g., the direction {circle around (1)} in FIG. **3B**) from the first housing and including a second area (e.g., the second area **R2** in FIG. **3B**) overlapping the first area in a slide-in state, an antenna structure (e.g., the antenna structure **213** in FIG. **3B**) disposed in the first housing to overlap the first area when the first housing is viewed from the top, a conductive part (e.g., the conductive part **223** in FIG. **3B**) disposed in the second area and electromagnetically connected to the antenna structure in the slide-in state, and wireless communication circuitry (e.g., the wireless communication module **192** in FIG. **1**) electrically connected to the antenna structure. The wireless communication circuitry may be configured to transmit and/or receive radio signals in at least one designated frequency band through the antenna structure and the conductive part in the slide-in state.

(81) According to various example embodiments, the wireless communication circuitry may be configured to transmit and/or receive radio signals in the at least one frequency band through the antenna structure in a slide-out state.

(82) According to various example embodiments, the antenna structure may include at least one conductive part divided through at least one non-conductive part in the first area.

(83) According to various example embodiments, the antenna structure may include a conductive

pattern disposed in an internal space and/or on an internal surface of the first housing. At least a part of the conductive pattern may be electrically connected to a conductive pad exposed to an external surface of the first housing in the first area.

(84) According to various example embodiments, the electronic device may further include a conductive connection member disposed to be electrically connected to the conductive pad in an internal space of the second housing. The conductive connection member may electrically connect the conductive part to the conductive pad in a second direction substantially perpendicular to the first direction in the slide-in state.

(85) According to various example embodiments, the conductive connection member may include at least one of a conductive spring, a conductive tape or a conductive C-clip.

(86) According to various example embodiments, the antenna structure may be disposed at a location to be capacitively coupled to the conductive part in the slide-in state.

(87) According to various example embodiments, the conductive part may be divided through at least one non-conductive part in the second area.

(88) According to various example embodiments, the conductive part may be disposed to have a length substantially in the same direction as the antenna structure in the second area.

(89) According to various example embodiments, the conductive part may be formed through a slit having a length substantially in the same direction as the antenna structure in the second area.

(90) According to various example embodiments, in the slide-in state, when the first housing is viewed from the top, the antenna structure may be disposed at a location at least partially overlapping the slit.

(91) According to various example embodiments, the first housing may include a first face, a second face facing a direction opposite a direction of the first face, and a first side member at least partially surrounding a first space between the first face and the second face. The second housing may include a third face facing the same direction as the first face, a fourth face facing a direction opposite the direction of the third face, and a second side member at least partially surrounding a second space between the third face and the fourth face.

(92) According to various example embodiments, at least a part of the antenna structure may be disposed in the second face. At least a part of the conductive part may be disposed in the fourth face.

(93) According to various example embodiments, the first side member may include a first support member at least partially extending into the first space. The second side may include a second support member at least partially extending into the second space. In the slide-in state, the first support member and the second support member may at least partially overlap.

(94) According to various example embodiments, the first area may be disposed in at least a part of the first support member. The second area may be disposed in at least a part of the second support member.

(95) According to various example embodiments, the antenna structure may include at least one conductive part divided through at least one non-conductive part in the first support.

(96) According to various example embodiments, the conductive part may include at least one conductive part divided through at least one non-conductive part in the second support member.

(97) According to various example embodiments, the electronic device may further include a bendable member connected to the first housing and received in an internal space of the second housing in a way to be not seen from the outside in the slide-in state, and a flexible display disposed to be supported by the first housing and the bendable member.

(98) According to various example embodiments, the bendable member may be configured to be slid out from the internal space to form a plane substantially identical with a plane of the first housing in the slide-out state.

(99) According to various example embodiments, the flexible display may be disposed to have a first display area in the slide-in state and to have a second display area greater than the first display

area in a slide-out state.

(100) While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents.

Claims

1. An electronic device comprising: a slidable housing including first portion, and a second portion coupled to be slidable in a first direction from the first portion; an antenna structure, comprising at least part of an antenna, disposed in the first portion; a slit disposed in the second portion through a conductive part, the slit at least partially overlapping the antenna structure in a slide-in state; and wireless communication circuitry electrically connected to the antenna structure, wherein the wireless communication circuitry is configured to transmit and/or receive signals in at least one frequency band through the antenna structure and the slit in the slide-in state.
2. The electronic device of claim 1, wherein the wireless communication circuitry is configured to transmit and/or receive signals in the at least one frequency band through the antenna structure in a slide-out state.
3. The electronic device of claim 1, wherein the antenna structure comprises at least one conductive part divided through at least one non-conductive part.
4. The electronic device of claim 1, wherein: the antenna structure comprises a conductive pattern disposed in an internal space and/or on an internal surface of the first portion, and at least a part of the conductive pattern is electrically connected to a conductive pad exposed to an external surface of the first portion in the first area.
5. The electronic device of claim 4, further comprising a conductive connection member disposed to be electrically connected to the conductive pad in an internal space of the second portion, wherein the conductive connection member is configured to electrically connect the conductive part to the conductive pad in a second direction substantially perpendicular to the first direction in the slide-in state.
6. The electronic device of claim 5, wherein the conductive connection member comprises at least one of a conductive spring, a conductive tape or a conductive C-clip.
7. The electronic device of claim 1, wherein the antenna structure is disposed at a location capacitively coupled to the conductive part in the slide-in state.
8. The electronic device of claim 1, wherein the conductive part is divided through at least one non-conductive part.
9. The electronic device of claim 1, wherein the conductive part has a length in a direction substantially identical with a direction of the antenna structure.
10. The electronic device of claim 1, wherein the slit has a length in a direction substantially identical with a direction of the antenna structure.
11. The electronic device of claim 10, wherein in the slide-in state, when the first portion is viewed from above, the antenna structure is disposed at a location at least partially overlapping the conductive part.
12. The electronic device of claim 1, wherein: the first portion comprises a first surface, a second surface facing a direction opposite a direction of the first surface, and a first lateral member at least partially surrounding a first space between the first surface and the second surface, and the second portion comprises a third surface facing a same direction as the first surface, a fourth surface facing a direction opposite the direction of the third surface, and a second lateral member at least partially surrounding a second space between the third surface and the fourth surface.
13. The electronic device of claim 12, wherein: at least a part of the antenna structure is disposed in

the second surface, and at least a part of the slit and at least a part of the conductive part are disposed in the fourth surface.

14. The electronic device of claim 12, wherein: the first lateral member comprises a first support member at least partially extending into the first space, the second lateral member comprises a second support member at least partially extending into the second space, and in the slide-in state, the first support member and the second support member at least partially overlap.

15. The electronic device of claim 14, wherein: the first portion comprises a first area and the second portion comprises a second area overlapping the first area in a slide-in state, the antenna structure is disposed in the first area, the slit and the conductive part are disposed in the second area, the first area is disposed in at least a part of the first support member, and the second area is disposed in at least a part of the second support member.

16. The electronic device of claim 15, wherein the antenna structure comprises at least one conductive part divided through at least one non-conductive part in the first support member.

17. The electronic device of claim 15, wherein the conductive part comprises at least one conductive part divided through at least one non-conductive part in the second support member.

18. The electronic device of claim 1, further comprising: a bendable member connected to the first portion and received in an internal space of the second portion to be not seen from an outside in the slide-in state; and a flexible display disposed to be supported by the first portion and the bendable member.

19. The electronic device of claim 18, wherein the bendable member is configured to be slid out from the internal space to form a plane substantially identical with a plane of the first portion in the slide-out state.

20. The electronic device of claim 19, wherein the flexible display is disposed to have a first display area in the slide-in state and to have a second display area greater than the first display area in a slide-out state.
